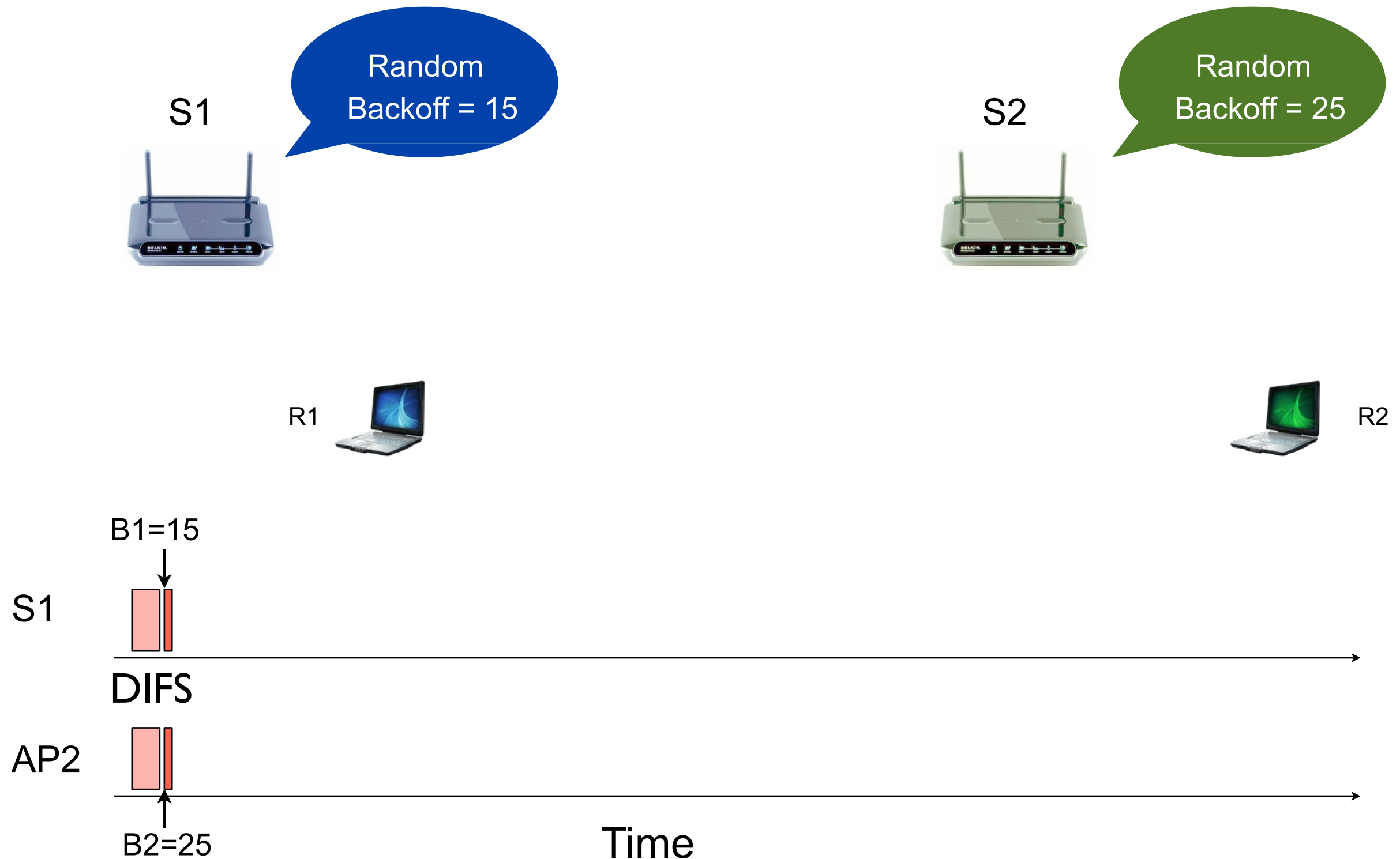


CSE 4415:: Wireless Networks

MAC Efficiency Boost :: Frequency-Domain Backoff

Lecture 7.1

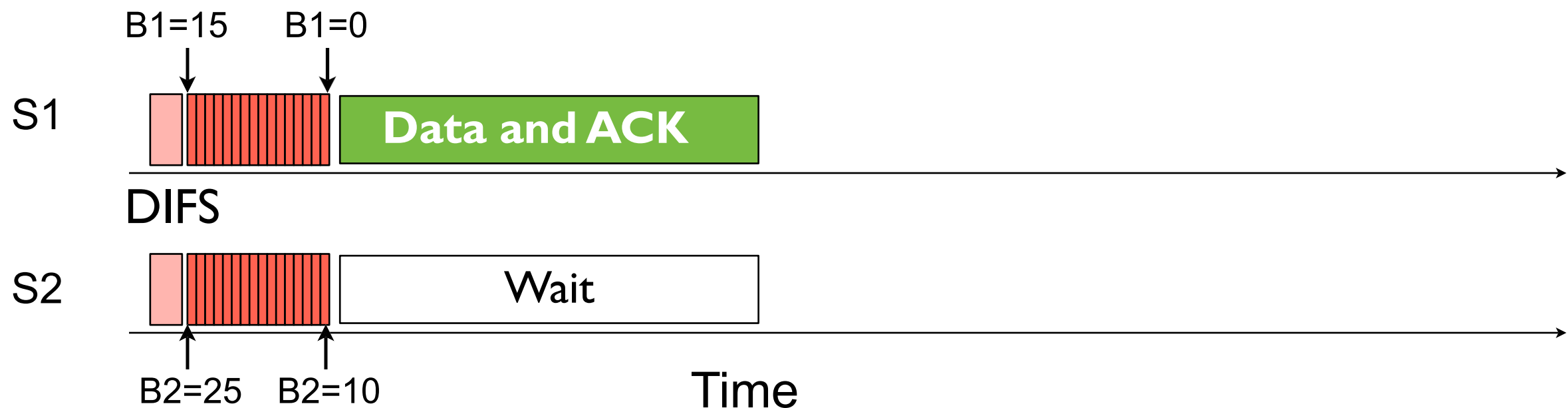
Current WiFi Channel Contention



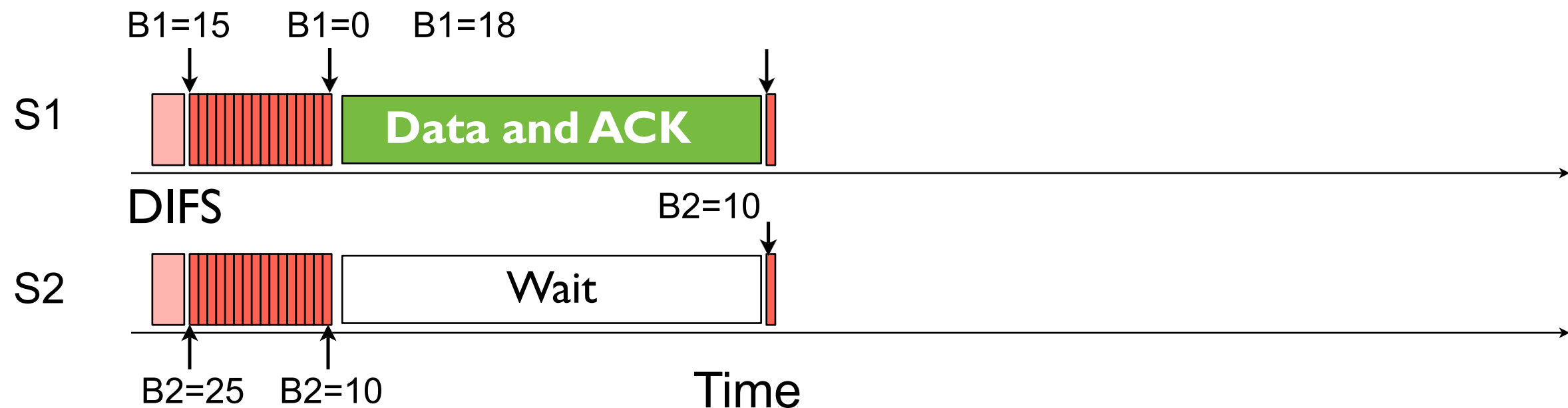
Current WiFi Channel Contention



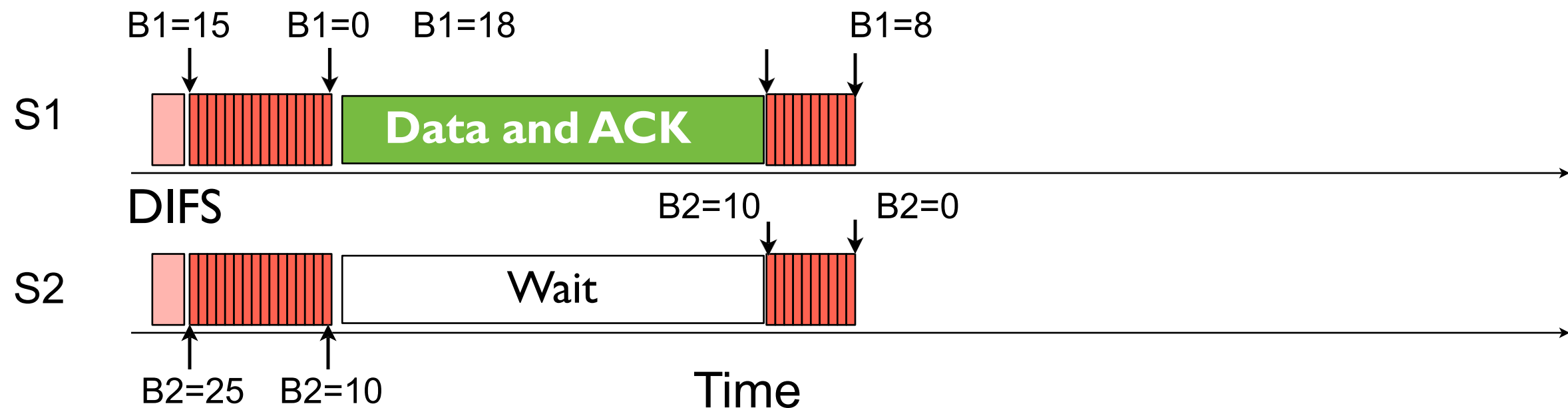
Current WiFi Channel Contention



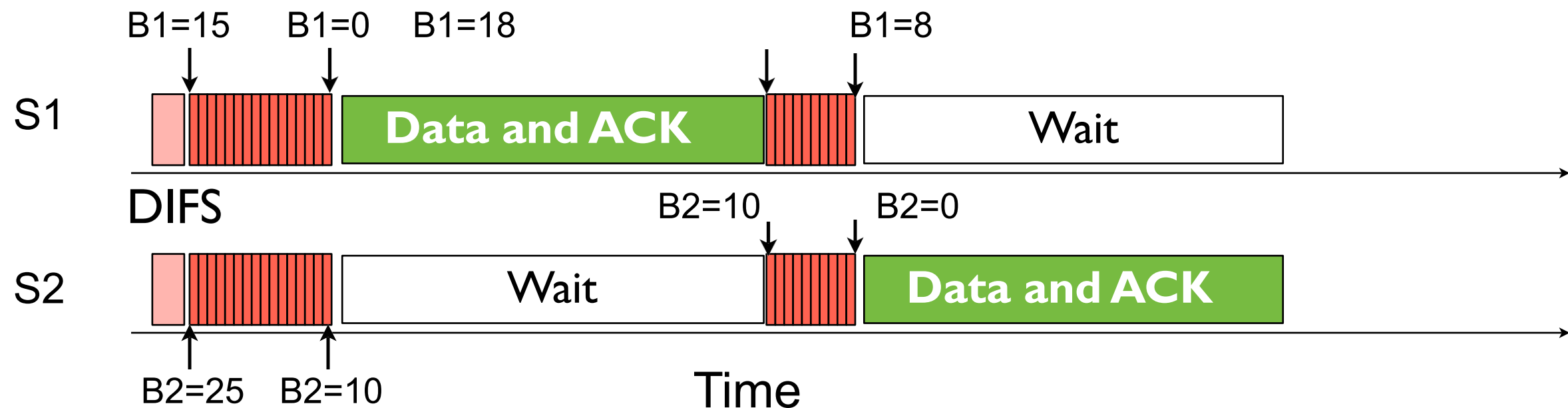
Current WiFi Channel Contention



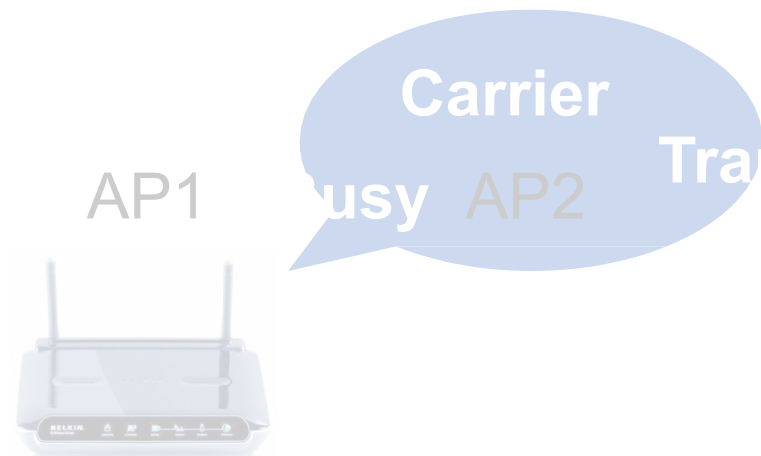
Current WiFi Channel Contention



Current WiFi Channel Contention



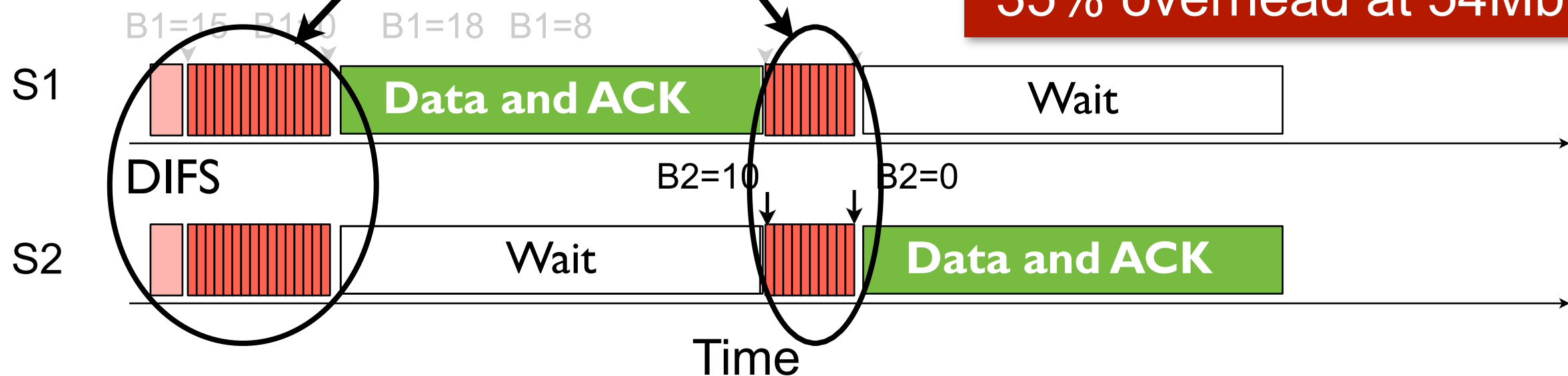
Current WiFi Channel Contention



High channel wastage due to backoff

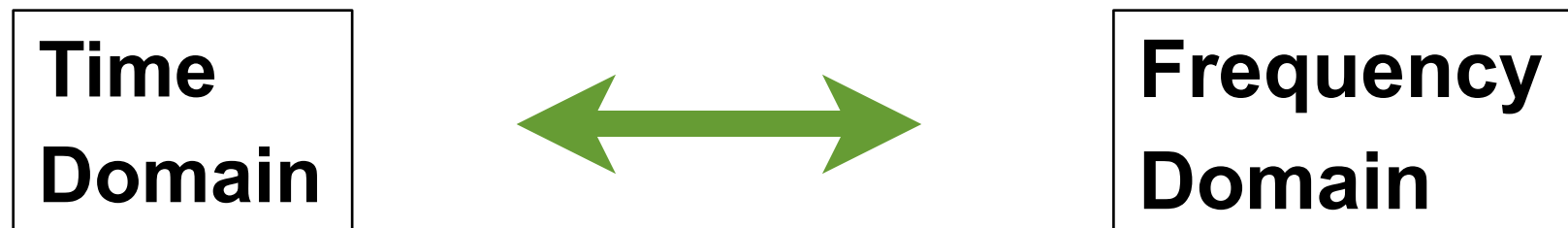


35% overhead at 54Mbps



Current WiFi Channel Contention

- Backoff is not fundamentally a time domain operation
 - ✂ Its implementation is in time domain

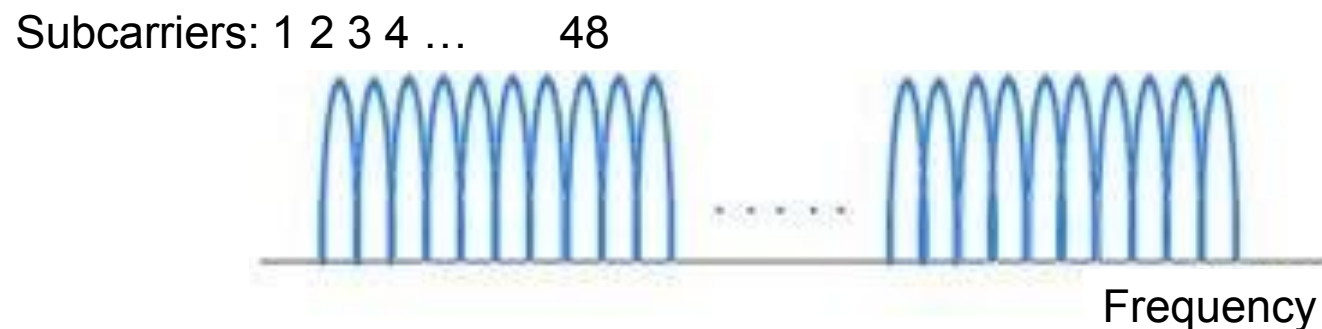


Can we implement backoff in frequency domain?
Are there any benefits in doing so?

Frequency domain contention resolution

□ 802.11 a/g/n PHY adopts OFDM

- ✂ Wideband channel divided into 48 narrowband subcarriers
- ✂ Copes better with fast, frequency selective fading
- ✂ Purely a PHY motivation

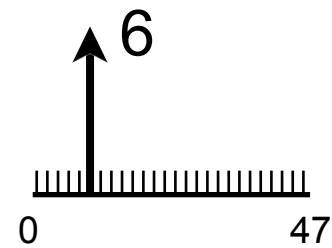


Proposed Protocol: Back2F

- MAC Opportunity: Pretend OFDM subcarriers as integers
- Emulate randomized backoff

Back2F: Main Idea

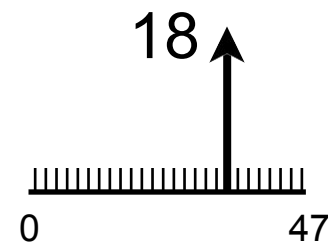
- Replace temporal with subcarrier transmission



S1



Backoff = 6



S2



Backoff = 18



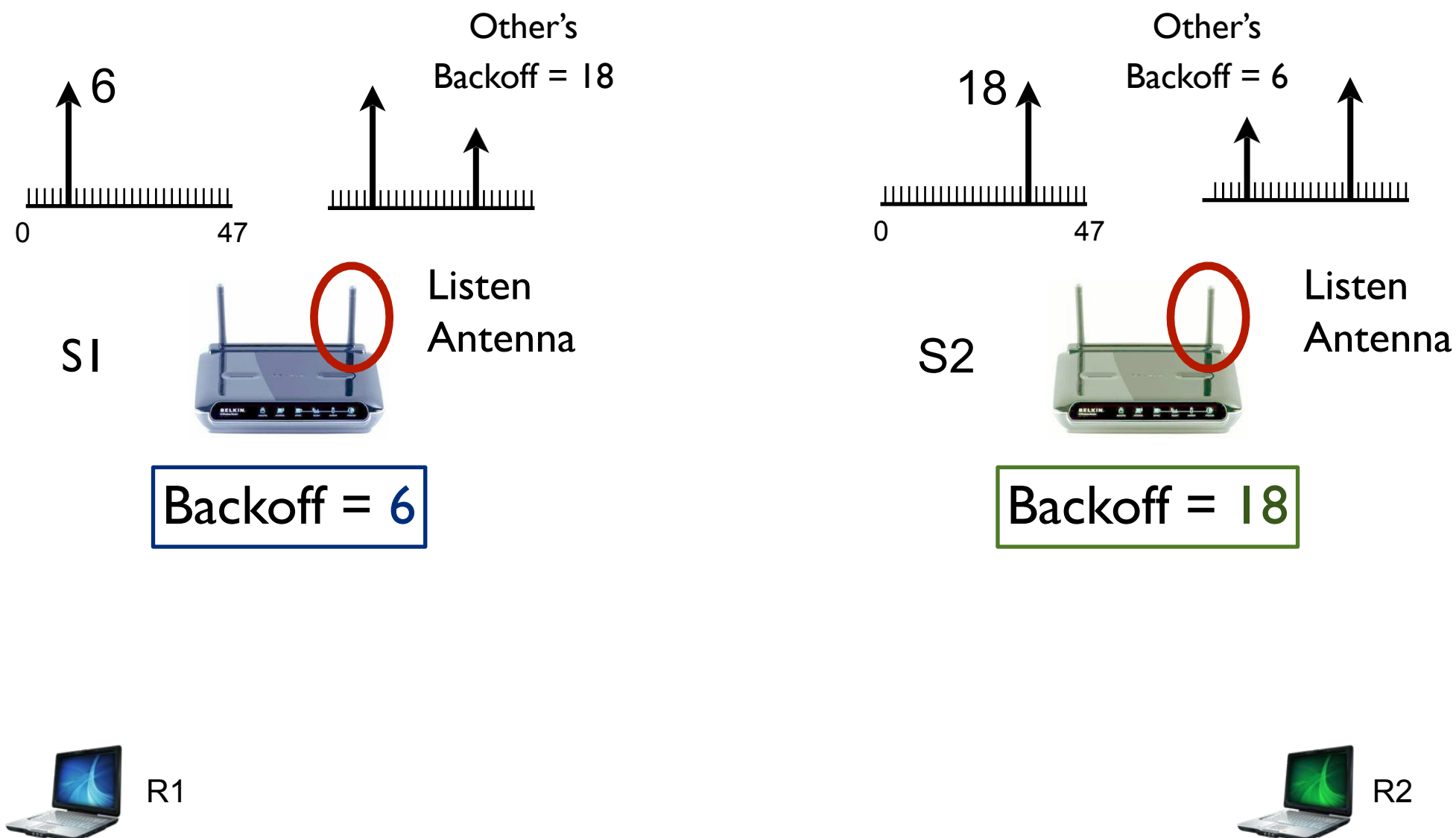
R1



R2

Back2F: Main Idea

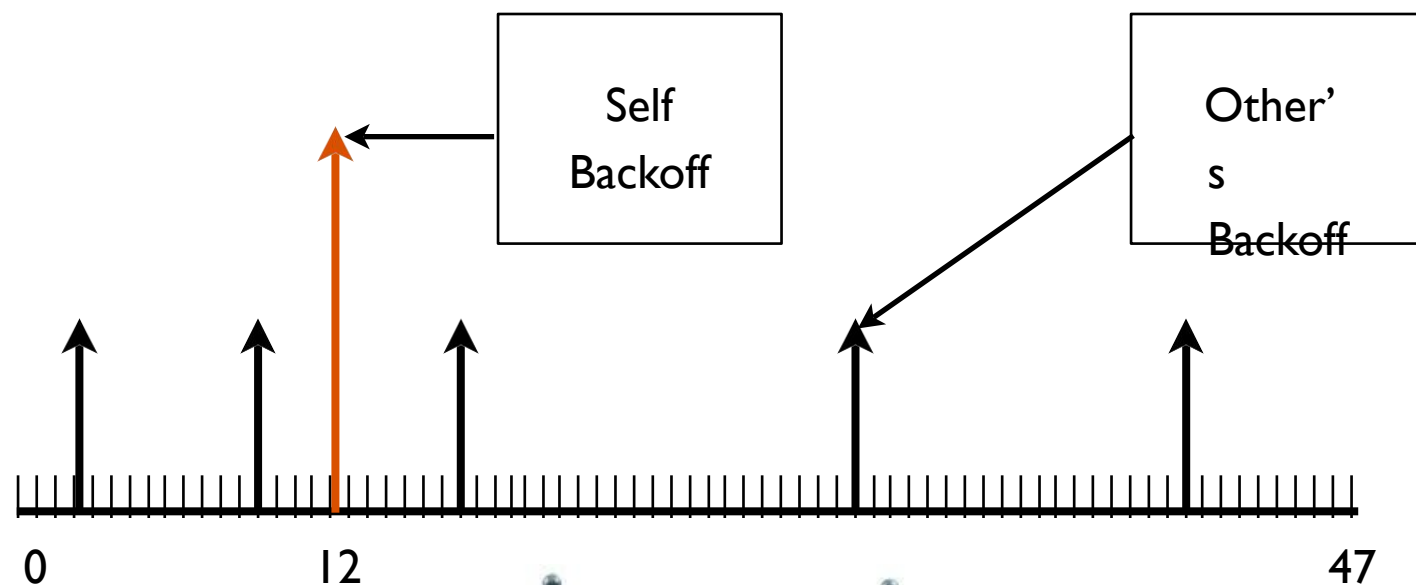
- Replace temporal with subcarrier transmission



Both Stations learn S1 is the winner

Back2F: Scheduled Transmission

- Active subcarriers imply backoff chosen by other Stations
 - ✂ Each AP knows its rank in the sequence
 - ✂ Enables back to back TDMA like transmission



API

Rank in TDMA: 3

Is there a benefit with frequency domain backoff?

-1500 bytes at 54Mbps ~ 250 micro sec.

-Avg. temporal backoff ~ **100 micro sec.**

Big overhead

Frequency backoff = 1 OFDM symbol = 4 micro sec

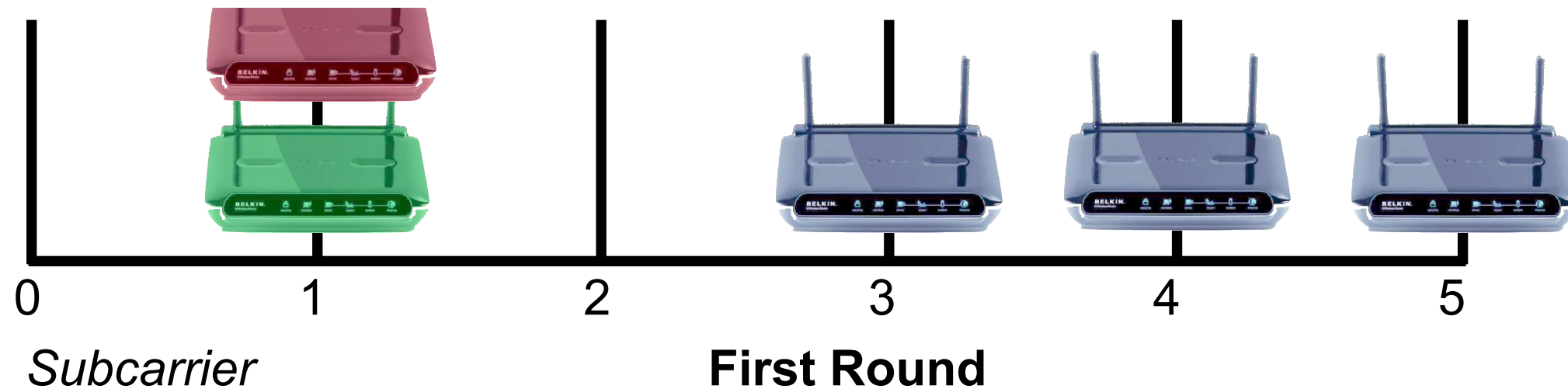
4 microsecond overhead ☐ dramatic performance enhancement relative to 100 microsecond overhead

- Instead of waiting **100 microseconds**, we only "wait" **4 microseconds** by using this smarter method (frequencies instead of time slots).
- This leads to **much faster channel access** and **higher throughput**.

Will Stations Collide During Contention?

□ Introduce a second round of contention

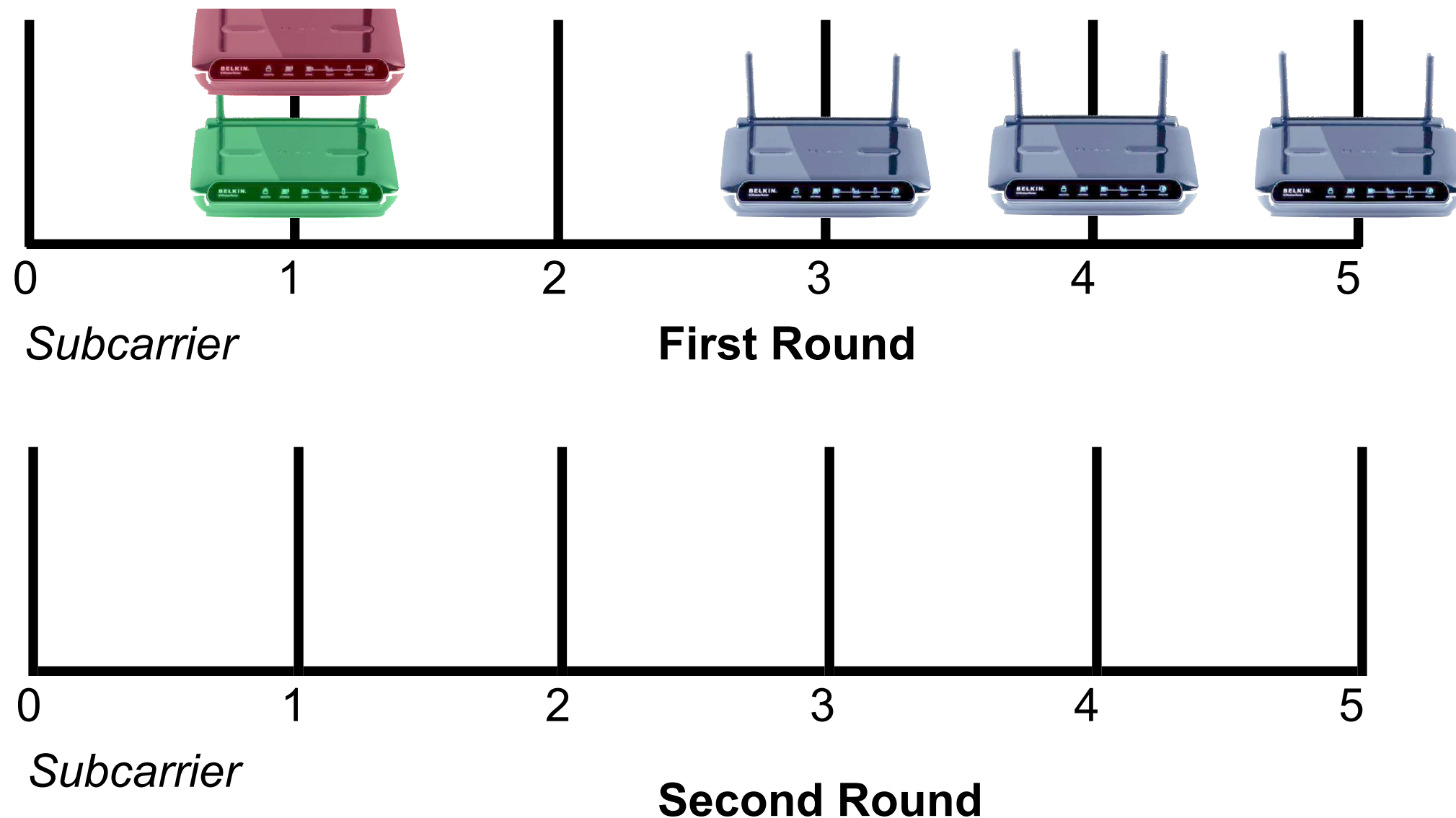
✂ Winners of first go to second



Will Stations Collide During Contention?

□ Introduce a second round of contention

✂ Winners of first go to second

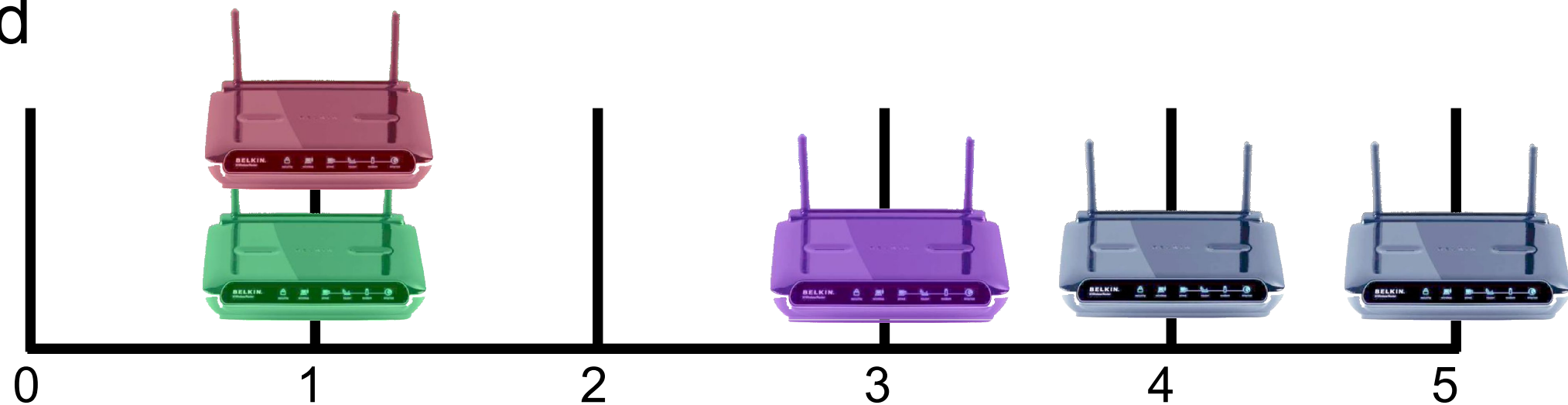


Only a Few Stations in Second Round?

TDMA will not be effective

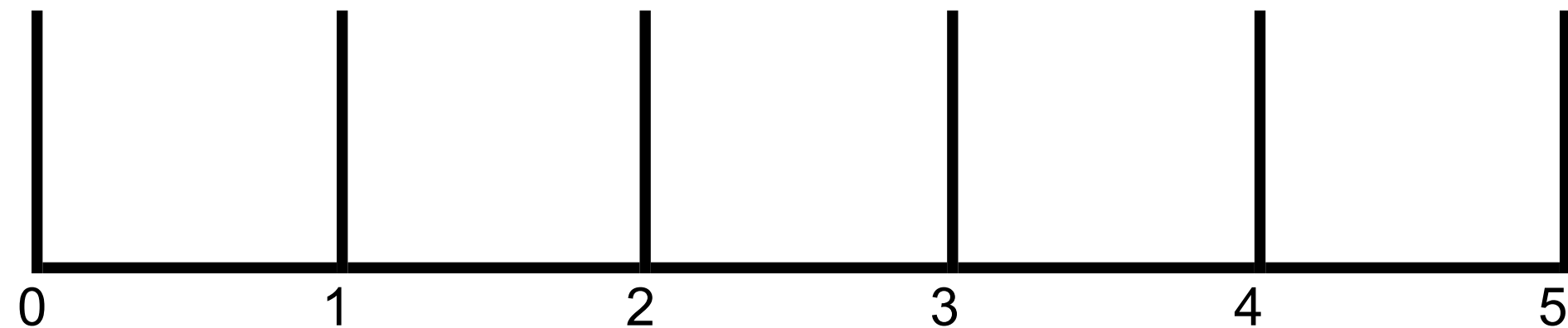
Optimize for TDMA

- Instead of only winners, a few more Stations to second round



Subcarrier

First Round

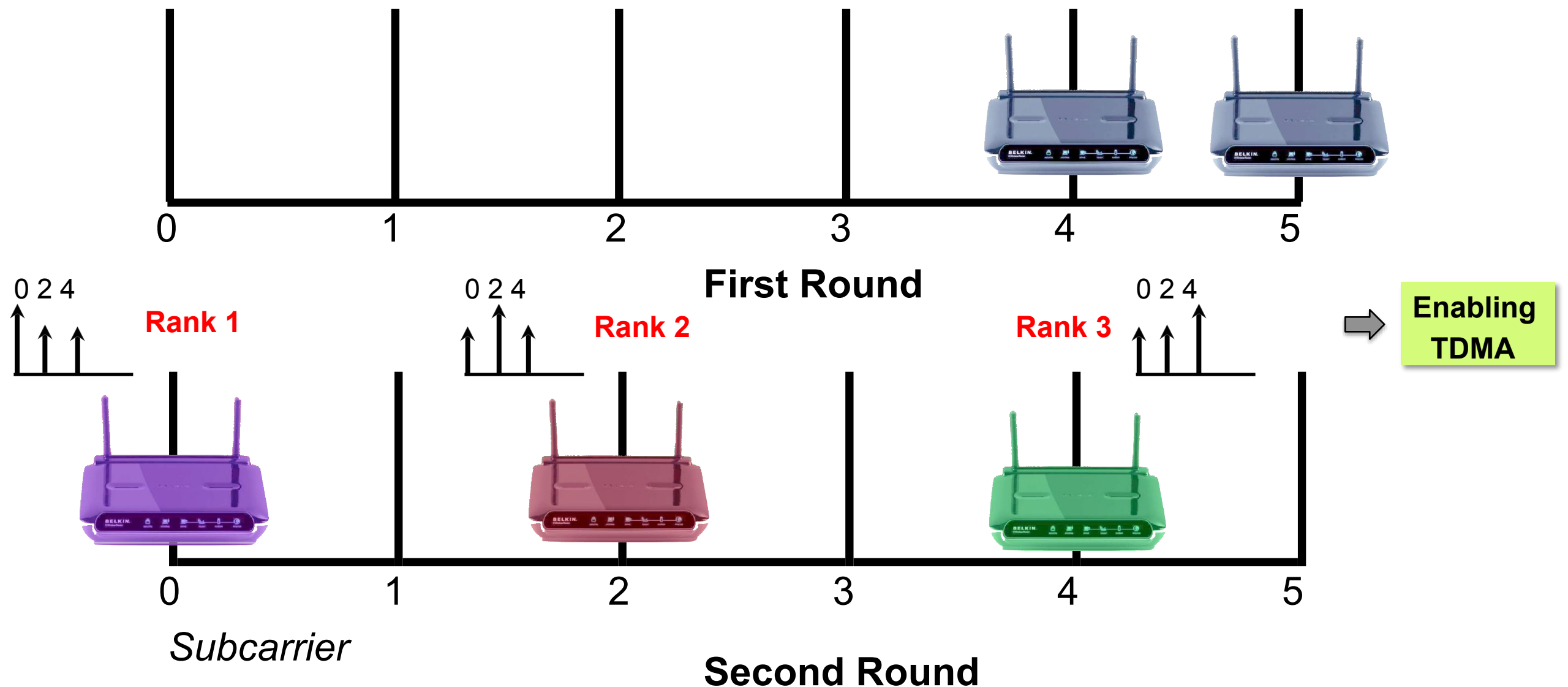


Subcarrier

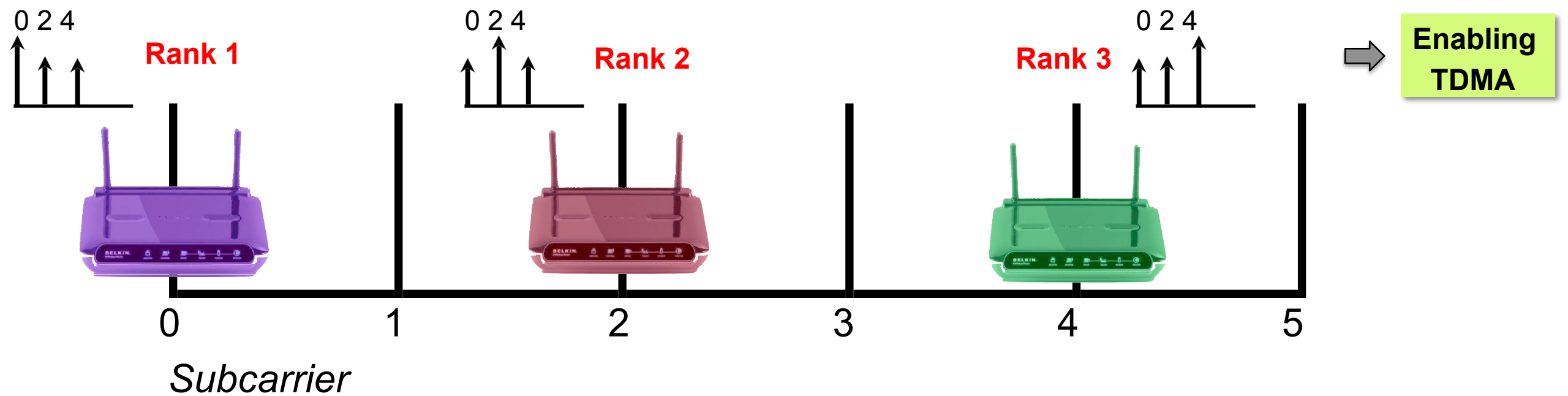
Second Round

Optimize for TDMA

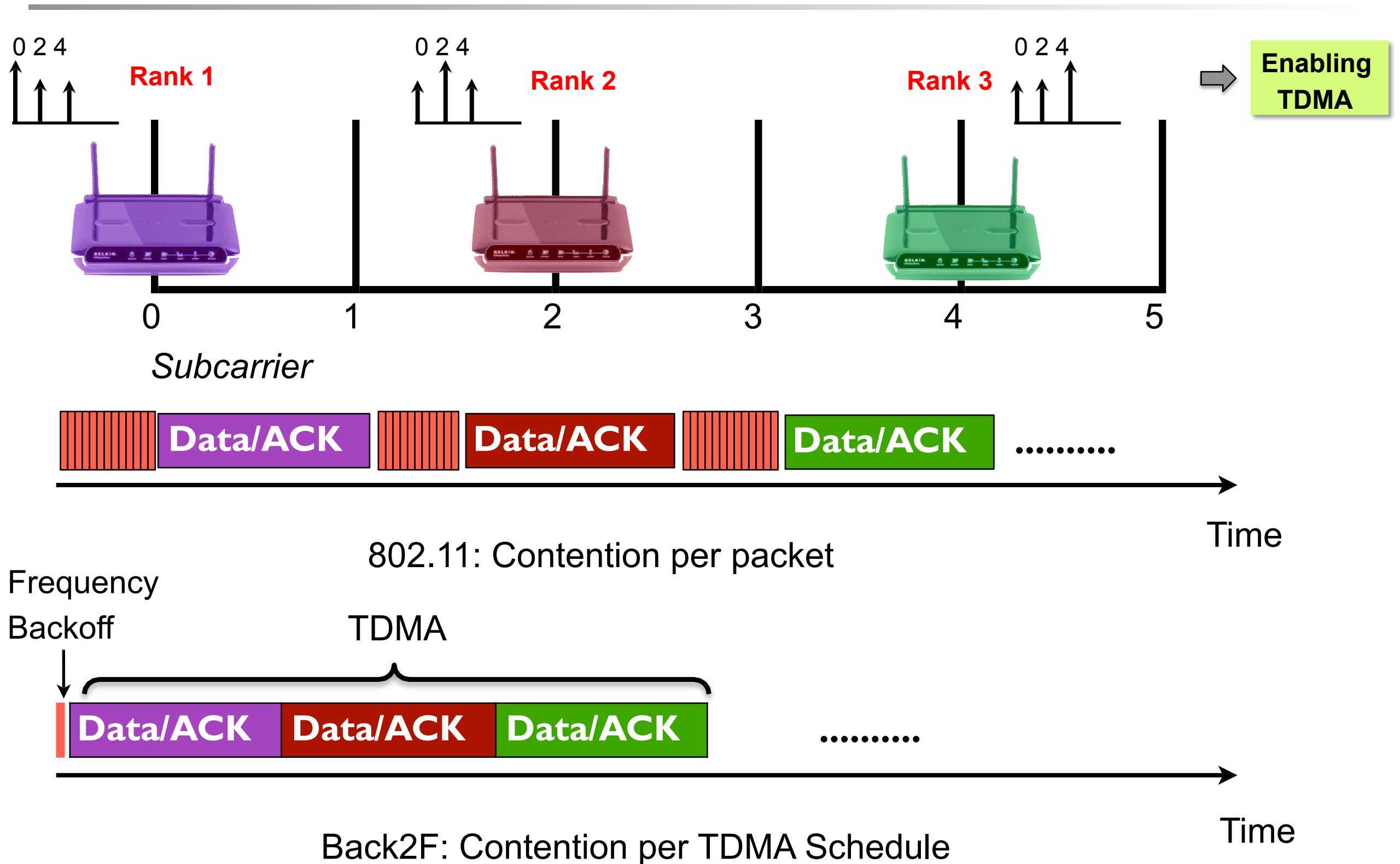
- Instead of only winners, a few more Stations to second round



Improved Channel Utilization



Improved Channel Utilization



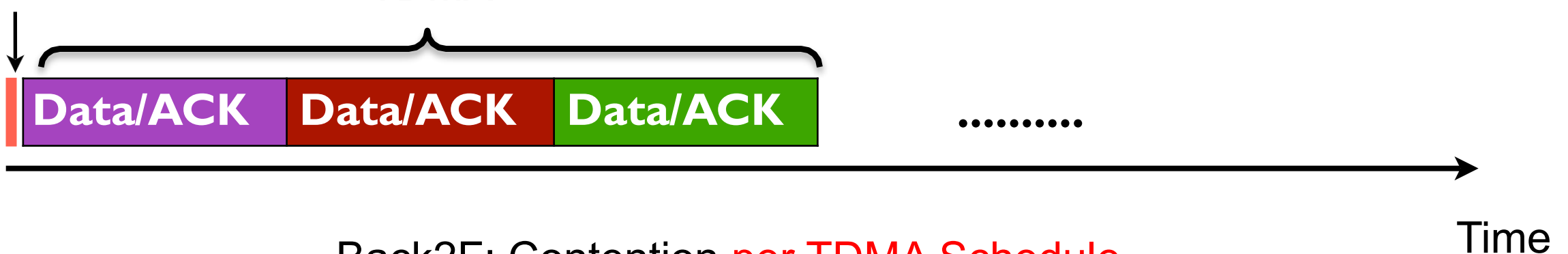
Improved Channel Utilization



802.11: Contention **per packet**

Frequency
Backoff

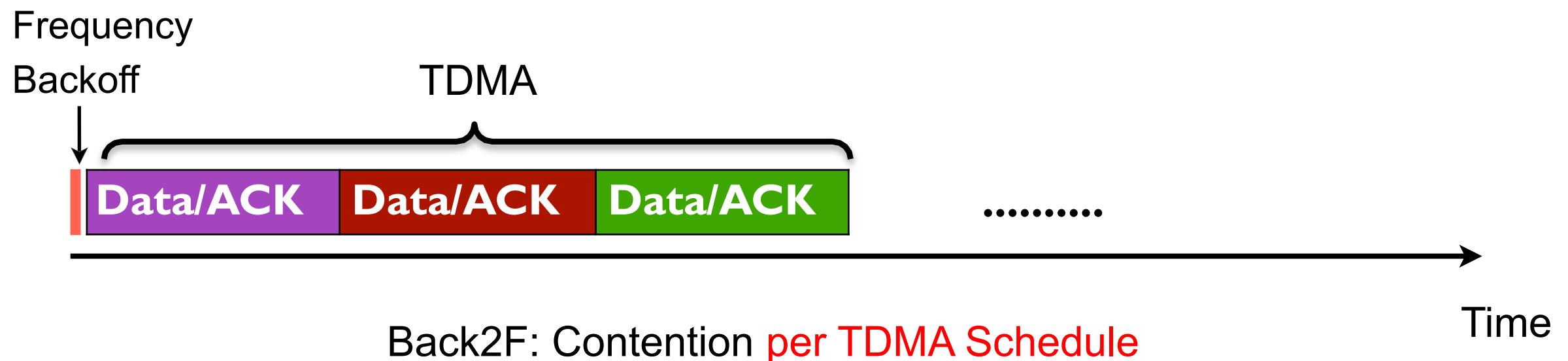
TDMA



Back2F: Contention **per TDMA Schedule**

Improved Channel Utilization

Once each station's rank is determined,
nodes can transmit in a **TDMA-like manner**
using the entire channel bandwidth—not just a specific subcarrier



Paper Summary

- Randomization is an effective method for contention resolution
- 802.11 time domain backoff requires channel to remain idle
- Observation: randomization possible in frequency domain
 - ✂ Using OFDM subcarriers
- Back2F: practical system realizing frequency domain contention
- Prevents collisions, provides upto 50% improvement in throughput

Paper Summary

- Multi-frequency assistance (OFDM technology based) protocol, creating Collision-free schedule.
- A slot assignment scheme where no slot remains idle.
- In this paper, they ask whether the entire backoff operation can be migrated to the frequency domain, thereby eliminating a long-standing source of channel inefficiency.
- Their core idea draws on OFDM subcarriers in modern WiFi radios, treating each subcarrier as an integer number.
- By transmitting a signal on a random subcarrier frequency, and using a second antenna to listen for all active frequencies, each node may be able to detect its rank among all contenders.
- Since signalling on subcarriers is almost instantaneous, the wastage from backoff can become negligible.

Paper Summary

- Once each station's rank is determined, nodes can transmit in a TDMA-like manner using the entire channel bandwidth—not just a specific subcarrier

Paper Summary: Advantage

- Fully Distributed

→ Works without any central controller or coordination messages between devices.

- Low Overhead

→ Doesn't need frequent control messages or feedback to function properly.

- Highly Efficient

→ Very few collisions and minimal wasted time; nodes can coordinate smoothly.

- No Idle Time Before Transmission

→ Unlike 802.11 which waits during backoff, Back2F doesn't waste time sensing an idle channel.

- Quick to Stabilize

→ Reaches an efficient working state quickly after nodes start contending.

Paper Summary: Limitation

- Needs Extra Antenna

→ Requires one antenna to transmit and another to listen at the same time, which may not be supported on current hardware.

- Sensitive to Channel Fluctuation

→ Subcarrier-based signaling can be affected by fading or interference in wireless environments.

- May Cause Unfairness with Legacy Wi-Fi (802.11)

→ Back2F might not treat older Wi-Fi devices fairly during mixed deployments.

Questions, comments?

Thank you